

Multi-Agent Single-Index Bandits

Learning unknown reward functions collaboratively under communication constraints

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From contextual bandits to an unknown reward function



A contextual bandit chooses an action x using its context and observes a noisy reward y .

Common models

$$\mathbb{E}[y \mid x] = \langle x, \theta_* \rangle \quad (\text{Linear}), \quad \mathbb{E}[y \mid x] = f(\langle x, \theta_* \rangle) \quad (\text{GLM}).$$

$$x \in \mathbb{R}^d \quad \longrightarrow \quad x^\top \theta_* \quad \longrightarrow \quad f(\cdot) \quad \longrightarrow \quad y$$

The gap

In GLMs, f is known. In many applications, the reward curve is unknown and may be non-monotone.

What if both θ_* and f are unknown?

Single-Index Bandits: the problem

Model

At round t , the learner observes K arms $\mathcal{X}_t = \{x_{t,a}\}_{a \in [K]}$, chooses x_t , and observes

$$y_t = f(\langle x_t, \theta_* \rangle) + \eta_t$$

where $\theta_* \in \mathbb{R}^d$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ are both unknown, with $\|\theta_*\|_1 = 1$.

$$x_t \in \mathbb{R}^d \xrightarrow{\theta_*} x_t^\top \theta_* \xrightarrow{f} y_t$$

Why is this hard?

If f is non-monotone,

$$x_1^\top \theta_* > x_2^\top \theta_* \not\Rightarrow f(x_1^\top \theta_*) > f(x_2^\top \theta_*).$$

The largest projection need not be the best arm.



Key idea: estimate θ_* without knowing f

Define the score function

$$S(x) = -\nabla \log p(x).$$

Under Stein's identity,

$$\mathbb{E}[yS(x)] = \mathbb{E}[f'(\langle x, \theta_* \rangle)] \theta_*.$$

Let

$$\mu_* = \mathbb{E}[f'(\langle x, \theta_* \rangle)].$$

Then

$$\boxed{\mathbb{E}[yS(x)] = \mu_* \theta_*}$$

so the direction of

$$\boxed{\hat{v} = \frac{1}{T_0} \sum_{t=1}^{T_0} y_t S(x_t)}$$

estimates the direction of θ_* without estimating f .

Key insight

Unknown f does not prevent us from estimating the index direction.



The SIB problem reduces to 1-D learning

Once θ_* is estimated, project each context onto the estimated direction:

$$z = x^\top \hat{\theta}.$$

$$x \in \mathbb{R}^d \xrightarrow{\hat{\theta}} z = x^\top \hat{\theta} \in \mathbb{R} \xrightarrow{f} f(z)$$

The original d -dimensional problem becomes learning an unknown function on a 1-D index.

$$[-1, 1] = I_1 \cup I_2 \cup \dots \cup I_M, \quad M = O(T^{1/3}).$$

ZoomSIB-UCB

The index domain is divided into bins, and a sleeping-UCB procedure learns which regions of the unknown reward curve are promising.



Setting

Consider N decentralized agents on a connected communication graph G . At each round t , each agent i observes an independent arm set, pulls $x_{i,t}$, and observes $y_{i,t} = f(\langle x_{i,t}, \theta_* \rangle) + \eta_{i,t}$. The parameter θ_* and link f are identical across agents but unknown. Agents can exchange messages with their neighbors subject to a communication budget.

Objective: Cumulative Network Regret

$$R_T(N) = \sum_{t=1}^T \sum_{i=1}^N [f(\langle x_{i,t}^*, \theta_* \rangle) - f(\langle x_{i,t}, \theta_* \rangle)].$$

Can decentralized collaboration improve per-agent regret when neither the index direction nor the link function is known?

Phase 1: Estimate θ_*

Each agent computes

$$\hat{\theta}_i \propto \frac{1}{T_0} \sum_t y_{i,t} S(x_{i,t}).$$

Then agents exchange their local estimates:

$$\hat{\theta} = \frac{1}{N} \sum_{i=1}^N \hat{\theta}_i$$

Exact pooled averaging
One d -dimensional message.

Phase 2: Learn f

Project:

$$z_{i,t} = x_{i,t}^\top \hat{\theta}.$$

Each agent maintains bin statistics

$$(n_j, S_j).$$

Agents periodically exchange these statistics. **Statistical pooling.**

Phase 1: exact pooling vs. Phase 2: statistical pooling



What do we want to find out?

1. Does collaboration help?

Compare per-agent regret under independent and federated learning:

$$R_T^{\text{federated}} \quad \text{vs.} \quad R_T^{\text{independent}}.$$

2. How much communication is enough?

Study the tradeoff

$$\text{Regret} \leftrightarrow \text{Communication}$$

using periodic, event-triggered, and quantized exchanges.

3. How does the network matter?

Study dependence on the number of agents, dimension, number of arms, and graph connectivity:

$$N, \quad d, \quad K, \quad 1 - |\lambda_2(\mathbf{P})|.$$

Dates	Work	Deliverable
29 Sep – 5 Oct	Reproduce ZOOMSIB-UCB and baselines: Random, LinUCB, ESTOR, GSTOR, IGP-UCB	Validated regret curves
6 – 18 Oct	Implement Fed-ZoomSIB: Stein averaging + cooperative sleeping-UCB	Federated speedup curves
19 – 23 Oct	Communication ablations: periodic sync, event-triggered updates, quantization	Regret vs. communication
24 – 29 Oct	One extension: safety or clustered index directions	Extension analysis
30 – 31 Oct	Benchmarking, final report, and presentation	Project submission
Metrics: Network regret Per-agent regret Messages / bytes		

Vary N , d , K , topology, and link shape

Where does our problem sit?



	Single Agent	Multi-Agent
Known reward structure	Linear / GLM Bandits	Cooperative Contextual Bandits
Unknown reward function	SIB / ZoomSIB-UCB	Our problem

Unknown link + Multi-agent cooperation = Federated SIB

Our focus is the interaction between statistical learning and communication constraints.

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Thank You!